



PROBABILITY THEORY

MTM2592-Gr:2

FOURTH WEEK

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VARIANCE AND STANDARD DEVIATION:

Definition: If X is a random variable with probability function $f(x)$ and expected value μ , then the variance of X is

$$V(X) = \text{Var}(X) = \sigma_X^2 = E[(X - \mu)^2] = \begin{cases} \sum_x (X - \mu)^2 f(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (X - \mu)^2 f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$

Theorem: The variance of a random variable X is

$$\text{Var}(X) = E(X^2) - \mu^2 \quad (\mu^2 = E^2(X))$$

Proof: $\text{Var}(X) = \sigma_X^2 = E[(X - \mu)^2] = \sum [x - \mu]^2 f(x)$

$$\sum_x [x - \mu]^2 f(x) = \sum_x x^2 f(x) - 2\mu \sum_x x \cdot f(x) + \mu^2 \sum_x f(x)$$

$$= E(X^2) - 2\mu^2 + \mu^2$$

$$= E(X^2) - \mu^2$$

$$\text{Var}(X) = \sigma_X^2 = E(X^2) - [E(X)]^2$$

Properties of Variance;

If c is a constant, X is a random variable,

1) $\text{Var}(c)=0$

Proof:
$$\text{Var}(c) = \frac{1}{k} \sum_{i=1}^k (c)^2 - \left[\frac{1}{k} \sum_{i=1}^k (c) \right]^2 = \frac{1}{k} \cdot k \cdot c^2 - \left(\frac{1}{k} \cdot k \cdot c \right)^2 = c^2 - c^2 = 0$$

2) $\text{Var}(X+c)=\text{Var}(X)$

Proof:
$$\begin{aligned} \text{Var}(X+c) &= \frac{1}{k} \sum_{i=1}^k (x_i + c)^2 - \left[\frac{1}{k} \sum_{i=1}^k (x_i + c) \right]^2 = \frac{1}{k} \sum_{i=1}^k (x_i^2 + 2c x_i + c^2) - (\bar{x} + c)^2 \\ &= \frac{1}{k} \sum_{i=1}^k x_i^2 + \frac{2c}{k} \sum_{i=1}^k x_i + \frac{1}{k} \cdot k \cdot c^2 - \bar{x}^2 - c^2 - 2c\bar{x} \\ &= \frac{1}{k} \sum_{i=1}^k x_i^2 - \bar{x}^2 = \text{Var}(X) \end{aligned}$$

3) $\text{Var}(cX)=c^2\text{Var}(X)$

Proof:
$$\begin{aligned} \text{Var}(cX) &= \frac{1}{k} \sum_{i=1}^k (cx_i)^2 - \left[\frac{1}{k} \sum_{i=1}^k (cx_i) \right]^2 = \frac{c^2}{k} \sum_{i=1}^k x_i^2 - (c\bar{x})^2 \\ &= \frac{c^2}{k} \sum_{i=1}^k x_i^2 - c^2 \bar{x}^2 = c^2 \text{Var}(X) \end{aligned}$$

4) $\text{Var}(aX+b)=a^2\text{Var}(X)$

Proof: Let, $Y=aX+b$, then

$$\begin{aligned}\text{Var}(Y) &= E\left[\left(Y - E(Y)\right)^2\right] = E\left[\left(aX + b - E(aX + b)\right)^2\right] \\ &= E\left[\left(aX + b - aE(X) - b\right)^2\right] \\ &= E\left[a^2 (X - E(X))^2\right] = a^2 E\left[(X - \mu)^2\right] = a^2 \text{Var}(X)\end{aligned}$$

Var is not a linear operator!

Definition: The positive square root of the variance is called the **standard deviation of X**.

$$\sigma_X = \sqrt{\text{Var}(X)}$$

Example:

$$f(x) = \begin{cases} \frac{3}{4}x(2-x), & \text{if } 0 < x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

a) Show that f is a probability density function.

b) Determine the cumulative distribution $F(x)$

c) Find the probabilities;

$$P(a < x \leq b) = F(b) - F(a)$$

$$P(X \leq 3/2), \quad P(X > 1/2), \quad P(0 < X \leq 7/4) \quad \text{and} \quad P(1/2 < X \leq 9/2)$$

Solution:

a) i) $f(x) \geq 0$ for all x .

$$ii) \int_{-\infty}^{\infty} f(x) dx = \int_0^2 \frac{3}{4}x(2-x) dx = \frac{3}{4} \left(x^2 - \frac{x^3}{3} \right) \Big|_0^2 = 1 \quad \checkmark$$

$$b) \quad F(x) = \begin{cases} 0, & x \leq 0 \\ \frac{3}{4} \left(x^2 - \frac{x^3}{3} \right), & 0 < x \leq 2 \\ 1, & x > 2 \end{cases}$$

$$F(x) = \int_{-\infty}^x f(u) du = \int_0^x \frac{3}{4}u(2-u) du = \frac{3}{4} \left(u^2 - \frac{u^3}{3} \right) \Big|_0^x = \frac{3}{4} \left(x^2 - \frac{x^3}{3} \right)$$

$$c) \quad P(x \leq 3/2) = F(3/2) = \frac{3}{4} \left(\left(\frac{3}{2} \right)^2 - \frac{\left(\frac{3}{2} \right)^3}{3} \right) =$$

$$P(x \leq x) = F(x) \quad P(x > 1/2) = 1 - P(x \leq 1/2) = 1 - F(1/2)$$

$$P(0 < x \leq 7/4) = F(7/4) - F(0)$$

Example: $f(x) = \begin{cases} k(2x+4), & \text{if } 0 < x \leq 6 \\ 0, & \text{elsewhere} \end{cases}$

$$P(X \leq x) = F(x)$$

- If f is a probability density function, find the value of k .
- Compute the mean $E(X) = \mu$ and standard deviation σ .
- Determine the cumulative function $F(x)$.
- Find the probabilities:

$$P(3 < X \leq 5), \quad P(X \geq 9/2), \quad P(|X-5| \leq 3), \quad P(X \leq 3 | 1 \leq X \leq 4) = ?$$

Solution:

$$\begin{aligned} P(|X-5| \leq 3) &= P(-3 \leq X-5 \leq 3) = P(2 \leq X \leq 8) \\ &= P(X=2) + P(2 < X \leq 8) = F(8) - F(2) \\ P(X \leq 3 | 1 \leq X \leq 4) &= \frac{P(1 \leq X \leq 3)}{P(1 \leq X \leq 4)} = \frac{F(3) - F(1)}{F(4) - F(1)} \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= \sigma_x^2 = E((X-\mu)^2) = E(X^2) - (E(X))^2 \\ E(X^2) &= \int_0^6 x^2(2x+4) dx = 15,6 \\ \text{Var}(X) &= 15,6 - (3,6)^2 = 2,64 \\ \sigma_x &= \sqrt{2,64} = 1,62 \end{aligned}$$

$$c) \quad \hat{F}(x) = \begin{cases} 0, & x \leq 0 \\ \frac{1}{60}(x^2+4x), & 0 < x \leq 6 \\ 1, & x > 6 \end{cases}$$

$$\hat{F}(x) = \frac{1}{60} \int_0^x (2u+4) du = \frac{1}{60} (x^2 + 4x)$$

Example:

$$f(x) = \begin{cases} \frac{3}{8}x, & \text{if } 0 < x \leq 2 \\ \frac{1}{12}, & \text{if } 2 < x \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

$$a) f(x) \geq 0$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\int_0^2 \frac{3}{8}x dx + \int_2^5 \frac{1}{12} dx = 1 \quad \checkmark$$

a) Show that $f(x)$ is a probability density function.

b) Find the cumulative distribution $F(x)$

c) Compute $P(1/4 \leq X \leq 5/2)$, and $P(|X| \geq 1)$

Solution:

b)

$$F(x) = \begin{cases} 0, & x \leq 0 \\ \frac{3}{16}x^2, & 0 < x \leq 2 \\ \frac{3}{4} + \frac{1}{12}(x-2), & 2 < x \leq 5 \\ 1, & x > 5 \end{cases}$$

$$F(x) = \int_{-\infty}^x f(u) du = \int_0^x \frac{3}{8}u du = \frac{3}{8} \frac{u^2}{2} \Big|_0^x = \frac{3}{16}x^2$$

$$F(2) = \frac{3}{16} \cdot 2^2 = \frac{3}{4}, \quad F(x) = \frac{3}{4} + \int_2^x \frac{1}{12} du = \frac{3}{4} + \frac{1}{12}(x-2)$$

$$P(x = \frac{1}{4}) + P(\frac{1}{4} < x \leq 5/2)$$

$$F(5/2) - F(1/4)$$

$$\begin{aligned} & P((x > 1) \cup (x \leq -1)) \\ &= P(x > 1) + P(x \leq -1) \\ &= 1 - P(x \leq 1) + P(x \leq -1) \\ &= 1 - F(1) + F(-1) \end{aligned}$$

$$= 1 - F(1) + F(-1) = 1 - \frac{3}{16} + 0 = \frac{13}{16}$$

MOMENTS

Definition: The r^{th} moment about the origin of a random variable X , denoted by m_r , is the expected value of $\underbrace{x^r}$; symbolically

$$E[(X - a)^r]$$

$$\underbrace{(m_r)} = E[\underbrace{(X)^r}] = \begin{cases} \sum_{i=1}^n \underbrace{(x_i)^r} \cdot \underbrace{f(x_i)} & \text{If } X \text{ is discrete} \\ \int_{-\infty}^{\infty} \underbrace{(x)^r} \cdot \underbrace{f(x)} \, dx & \text{If } X \text{ is continuous} \end{cases}$$

where $r=0,1,2,\dots$

Definition: The expected value of $(X-a)^r$, namely $E[(X-a)^r]$ is called the r^{th} moment about **a** of X .

$$\mu_r = E[\underbrace{(X - a)^r}] = \begin{cases} \sum_{i=1}^n \underbrace{(x_i - a)^r} \cdot f(x_i) & \text{If } X \text{ is discrete} \\ \int_{-\infty}^{\infty} \underbrace{(x - a)^r} \cdot f(x) \, dx & \text{If } X \text{ is continuous} \end{cases}$$

In order to exist the moments, the sum and the integral in the formula must be defined.

$$\underline{r=0} \Rightarrow E(X^0) = 1 = m_0$$

$$r=1 \Rightarrow E(X) = \mu = m_1 \quad \checkmark \quad E((\underset{\downarrow \mu}{x-a})^r)$$

$$r \Rightarrow, E(X^r) = m_r$$

If we take $a=\mu$, then $E[(X-\mu)^r]$ is called r-th moment about the mean of X. We denoted it by μ_r

$$\mu_r = E[(X-\mu)^r] = E\left[\binom{r}{0}x^r - \binom{r}{1}x^{r-1}\mu + \binom{r}{2}x^{r-2}\mu^2 + \dots + (-1)^k \binom{r}{k}x^{r-k}\mu^k + \dots + (-1)^r \binom{r}{r}\mu^r\right]$$

$$\downarrow = \binom{r}{0}E(x^r) - \binom{r}{1}E(x^{r-1})\mu + \dots + (-1)^r \binom{r}{r}\mu^r \quad \begin{matrix} E(a) = a \\ E(aX+b) = aE(x) + b \end{matrix}$$

If $a = \mu$ then

$$r=0 \Rightarrow \mu_0 = E[(X-\mu)^0] = 1$$

$$r=1 \Rightarrow \mu_1 = E[(X-\mu)] = 0 \quad \underbrace{E(x) - E(\mu)} = \mu - \mu$$

$$r=2 \Rightarrow \mu_2 = E[(X-\mu)^2] = \text{Var}(X)$$

Moments are called central moments according to the expected value(mean), moments according to point 0 are called ordinary moments.

$$\text{If } E[(X - \mu)^r] = \mu_r \text{ and } E[(X)^r] = m_r ;$$

Moments of the first three order

$$\mu_1 = E[X - E(X)] = E(X - m_1) = E(X) - m_1 = m_1 - m_1 = 0; \quad E(m_1) = m_1^2$$

$$\mu_2 = E[X - E(X)]^2 = E[(X - m_1)^2] = E(X^2) - 2m_1 E(X) + m_1^2 = m_2 - m_1^2;$$

$$\mu_3 = E[X - E(X)]^3 = E[(X - m_1)^3] = E(X^3) - 3m_1 E(X^2) + 3m_1^2 E(X) - m_1^3 = m_3 - 3m_1 m_2 + 2m_1^3;$$

Example: Find the first 3 moments of the numbers 2,4,7,9,10,11 about 0.

Solution:

$$E\left((X-\mu)^r\right) \quad r=1, 2, 3$$

$E(X)$
 $E(X^2)$
 $E(X^3)$

$$m_1 = E[(X)^1] = 2\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 7\left(\frac{1}{6}\right) + 9\left(\frac{1}{6}\right) + 10\left(\frac{1}{6}\right) + 11\left(\frac{1}{6}\right) = 7,17 \approx 7,2 \quad \checkmark$$

$$m_2 = E[(X)^2] = 2^2\left(\frac{1}{6}\right) + 4^2\left(\frac{1}{6}\right) + 7^2\left(\frac{1}{6}\right) + 9^2\left(\frac{1}{6}\right) + 10^2\left(\frac{1}{6}\right) + 11^2\left(\frac{1}{6}\right) = 61,8 \quad \checkmark$$

$$m_3 = E[(X)^3] = 2^3\left(\frac{1}{6}\right) + 4^3\left(\frac{1}{6}\right) + 7^3\left(\frac{1}{6}\right) + 9^3\left(\frac{1}{6}\right) + 10^3\left(\frac{1}{6}\right) + 11^3\left(\frac{1}{6}\right) = 579,2 \quad \checkmark$$

$\sum (x)^r f(x)$

Example Find the first 3 moments of the numbers 2,4,7,9,10,11 about mean.

Solution:

$$\mu_1 = E[(X-\mu)] = 0 \quad \Rightarrow \quad E(X) - E(\mu) = 0 \quad \mu = \frac{2+4+7+9+10+11}{6} = 7,166 \approx 7,2$$

$$\mu_2 = E[(X-\mu)^2] = (2-7,2)^2 \frac{1}{6} + (4-7,2)^2 \frac{1}{6} + (7-7,2)^2 \frac{1}{6} + (9-7,2)^2 \frac{1}{6} + (10-7,2)^2 \frac{1}{6} + (11-7,2)^2 \frac{1}{6} = 10,47 \quad \checkmark$$

$$\mu_3 = E[(X-\mu)^3] = (2-7,2)^3 \frac{1}{6} + (4-7,2)^3 \frac{1}{6} + (7-7,2)^3 \frac{1}{6} + (9-7,2)^3 \frac{1}{6} + (10-7,2)^3 \frac{1}{6} + (11-7,2)^3 \frac{1}{6} = -15,12 \quad \checkmark$$

MOMENT GENERATING FUNCTIONS

Definition: The moment-generating function of a random variable X is given by

$$\underline{M_X(t)} = E[\underline{e^{tX}}] = \begin{cases} \sum_{i=1}^k \underline{e^{x_i t}} \cdot \underline{f(x_i)} & \text{if } X \text{ is } \underline{\text{discrete}} \\ \int_{-\infty}^{\infty} \underline{(e^{xt})} \cdot \underline{f(x)} dx & \text{if } X \text{ is } \underline{\text{continuous}} \end{cases}$$

Remark: By using Maclaurin's Series expansion, we have

$$e^{tx} = 1 + tx + \frac{t^2 x^2}{2!} + \frac{t^3 x^3}{3!} + \dots + \frac{t^r x^r}{r!} + \dots \quad \checkmark$$

Then,

$$M_X(t) = E(e^{tx}) = 1 + tE(x) + \frac{t^2}{2!} E(x^2) + \frac{t^3}{3!} E(x^3) + \dots + \frac{t^r}{r!} E(x^r) + \dots$$

$$\left. \frac{dM_X(t)}{dt} \right|_{t=0} = \cancel{m_1} + \cancel{\frac{2t}{2!} m_2} + \cancel{\frac{3t^2}{3!} m_3} + \dots$$

$$1 + tm_1 + \frac{t^2}{2!} m_2 + \frac{t^3}{3!} m_3 + \dots + \frac{t^r}{r!} m_r + \dots$$

Properties of Moment Generating Function

1) * If $t = 0$ is put in the 1st derivative of the moment generating function

$$M_x'(t = 0) = \left. \frac{dM_x(t)}{dt} \right|_{t=0} = E(\underline{X}) = \underline{\mu} = \underline{m_1}$$

$E(X)$ is the first moment of X random variable.

* If $t = 0$ is put in the 2nd derivative of the moment generating function

$$M_x''(t = 0) = \left. \frac{d^2 M_x(t)}{dt^2} \right|_{t=0} = E(\underbrace{X^2}) = \mu_2$$

$E(X^2)$ is the second moment of X random variable

* If $t = 0$ is put in the n^{th} derivative of the moment generating function

$$M_x^{(n)}(t = 0) = \left. \frac{d^n M_x(t)}{dt^n} \right|_{t=0} = E(X^n) = \mu_n$$

$E(X^n)$ is the n^{th} moment of X random variable.

Note:
$$\text{Var}(X) = E(X^2) - [E(X)]^2 = M_x''(0) - [M_x'(0)]^2$$

2) If a and b are constants, then

i)
$$M_{X+a}(t) = E\left[e^{(X+a)t}\right] = e^{at} M_X(t)$$

ii)
$$M_{bX}(t) = E\left[e^{(bX)t}\right] = M_X(bt)$$

$$M_X(t) = E(e^{tX})$$

iii) If $Y=aX+b$, then

$$M_Y(t) = E(e^{tY}) = E[e^{t(aX+b)}] = e^{tb} \underbrace{E(e^{aXt})}_{M_X(at)} = e^{tb} M_X(at)$$

iv)
$$M_{\frac{X+a}{b}}(t) = E\left[e^{\left(\frac{X+a}{b}\right)t}\right] = e^{\frac{a}{b}t} M_X\left(\frac{t}{b}\right)$$

$$M_X(at)$$

3) The moment generating function is unique. This means that if two random variables have the same moment-generating function, they have the same probability distribution. If they have different moment generating functions, they can be said to have different distributions.

Example: Given that X has the probability function $f(x) = \frac{1}{8} \binom{3}{x}$

for $x=0,1,2,3$, find the moment generating function of the random variable X and use it to determine m_1 and m_2 . $\rightarrow E(x^2)$

< Discr ?
Cont ?

Solution:

$$M_X(t) = E(e^{tX}) = \frac{1}{8} \cdot \sum_{x=0}^3 e^{tx} \binom{3}{x}$$

$$\sum_x e^{tx} \cdot f(x)$$

$$= \frac{1}{8} (1 + 3e^t + 3e^{2t} + e^{3t})$$

$$= \frac{1}{8} (1 + e^t)^3 \quad \checkmark$$

$$Var(X) = E(X^2) - (E(X))^2$$

$$m_1 = M'_X(0) = \frac{3}{8} (1 + e^t)^2 e^t \Big|_{t=0} = \frac{3}{2} \quad \checkmark \quad E(X)$$

$$m_2 = M''_X(0) = \frac{3}{4} (1 + e^t) e^{2t} + \frac{3}{8} (1 + e^t)^2 e^t \Big|_{t=0} = 3 \quad E(X^2)$$

Example: Find the moment-generating function of the random variable X whose probability function is

$$f(x) = \begin{cases} \frac{1}{2} e^{-x/2}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

and use it to find the mean and the standard deviation.

Solution:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} \cdot \underbrace{f(x)}_{\substack{\text{probability} \\ \text{function}}} dx = \int_0^{\infty} e^{tx} \cdot \frac{1}{2} e^{-x/2} dx = \frac{1}{2} \int_0^{\infty} e^{x(t-1/2)} dx$$

$$= \frac{1}{2} \cdot \frac{1}{(t-1/2)} e^{x(t-1/2)} \Big|_0^{\infty} = \frac{1}{(2t-1)} (0-1) = \frac{1}{1-2t} \quad \boxed{t < 1/2}$$

$$E(X) = \mu = m_1 = M'_X(t=0) = \frac{dM_X(t)}{dt} \Big|_{t=0} = - (1-2t)^{-2} \cdot (-2) = \frac{2}{(1-2t)^2} \Big|_{t=0} = 2$$

$$\sigma_x = \sqrt{(E(X^2) - \mu^2)}$$

$$E(X^2) = m_2 = M''_X(t=0) = \frac{d^2 M_X(t)}{dt^2} \Big|_{t=0} = \frac{8}{(1-2t)^3} \Big|_{t=0} = 8$$

$$\sigma_x^2 = 8 - 2^2 = 4, \quad \sigma_x = \sqrt{4} = 2$$

WORKSHEET

1) Let A and B are any events. If

$$P(A) = 1/2, \quad P(B) = 1/3, \quad P(A \cap B) = 1/4$$

then compute the probabilities given below.

$$\text{i) } P(A|\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})} = \frac{1/4}{1/3} = 3/4$$

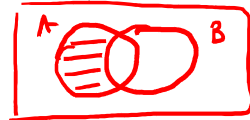
$$\text{ii) } P(\bar{B}|A) = \frac{P(\bar{B} \cap A)}{P(A)} = \frac{1/4}{1/2} = 1/2$$

$$\text{iii) } P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$$

$$\text{iv) } P(\bar{A}|\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{1 - P(A \cup B)}{1 - P(B)} = \frac{1 - 7/12}{1 - 1/3}$$

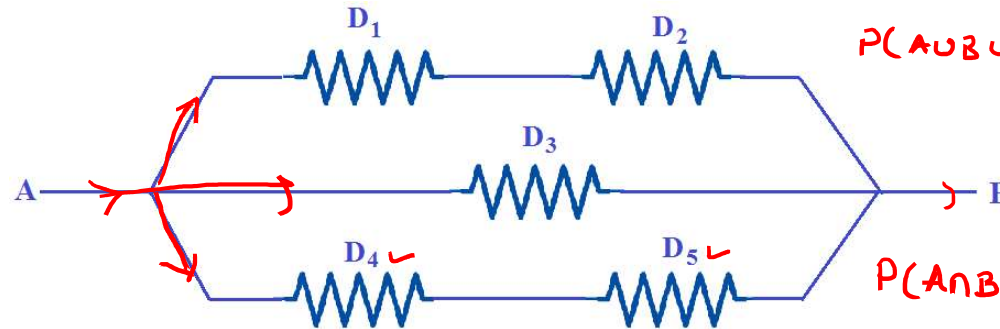
$$\text{v) } P(\bar{B}|\bar{A}) = \frac{P(\bar{B} \cap \bar{A})}{P(\bar{A})}$$

$$\text{vi) } P(A \cap \bar{B}) = P(A - B) = P(A) - P(A \cap B)$$



$$= \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

2) It is known that the current passes if the electrical switches are closed in an electrical circuit. In the following system, each of switches operates independently from each other. If the probability of switches being closed is 0.7, what is the probability that the current passes from A to B?



$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$P(A \cap B) = P(A) \cdot P(B)$$

$K : \{ \text{The current passes from A to B} \}$

$D_i = \{ \text{The switch } D_i \text{ is closed} \}$

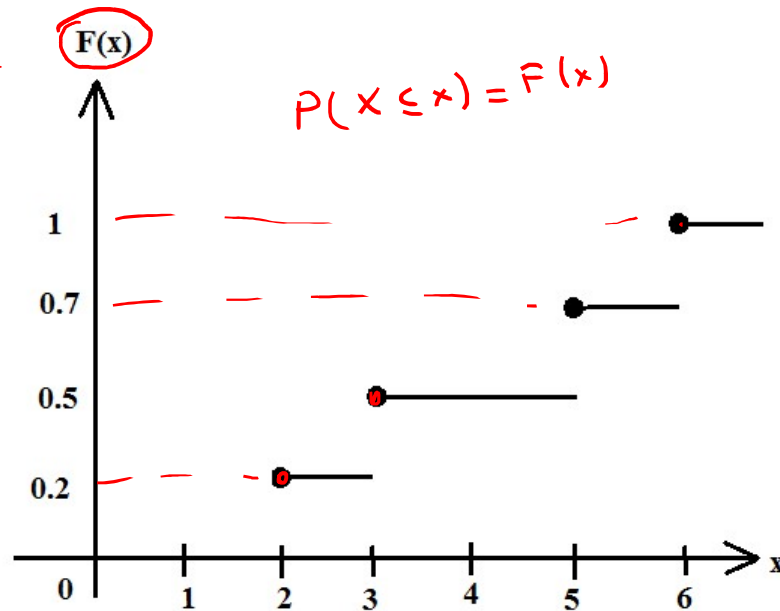
$$K = (\overline{D_1 \cap D_2}) \cup \overline{D_3} \cup (\overline{D_4 \cap D_5})$$

$$P(K) = P(D_1 \cap D_2) + P(D_3) + P(D_4 \cap D_5) - P(D_1 \cap D_2 \cap D_3) - P(D_1 \cap D_2 \cap D_4 \cap D_5) - P(D_3 \cap D_4 \cap D_5) + P(D_1 \cap D_2 \cap D_3 \cap D_4 \cap D_5)$$

$$P(K) = P(D_1)P(D_2) + P(D_3) + P(D_4)P(D_5) - P(D_1)P(D_2)P(D_3) - P(D_1)P(D_2)P(D_3)P(D_4)P(D_5) - P(D_3)P(D_4)P(D_5) + P(D_1)P(D_2)P(D_3)P(D_4)P(D_5)$$

$$= (0.7)^2 + 0.7 + (0.7)^2 - (0.7)^3 - (0.7)^4 - (0.7)^3 + (0.7)^5 = 0.922$$

3) By using graph of cumulative distribution $F(x)$, find the probabilities;



i) $P(X < 2) = 0$

ii) $P(X \leq 2) = F(2) = 0.2$

iii) $P(X > 3) = 1 - P(X \leq 3) = 1 - F(3) = 1 - 0.5 = 0.5$

iv) $P(X \geq 3) = 1 - P(X < 3) = 1 - F(2) = 1 - 0.2 = 0.8$

v) $P(X = 5)$

vi) $P(X = 6)$

v) $P(X = 5) = P(X \leq 5) - P(X < 5) = F(5) - 0.5 = 0.7 - 0.5 = 0.2$

vi) $P(X = 6) = P(X \leq 6) - P(X < 6) = F(6) - 0.7 = 1 - 0.7 = 0.3$

4) If $f(x)$ is a probability function.

$$f(x) = \begin{cases} c(x+1), & 1 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$$

- Find the value of c .
- Determine the cumulative func.
- Find $P(1 \leq x \leq 2)$

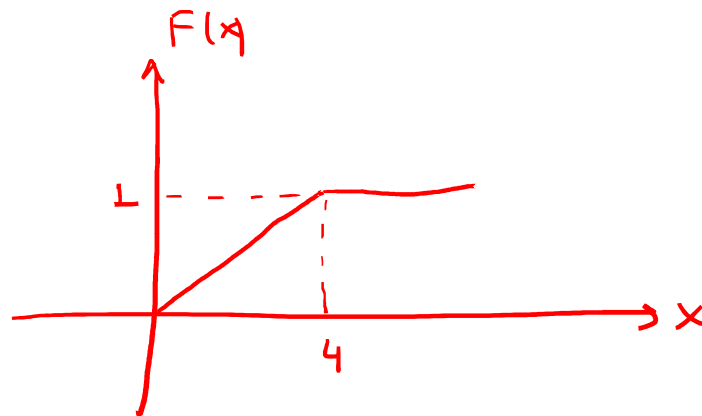
$$c = 1/6$$

$$F(x) = \begin{cases} 0, & x \leq 1 \\ \int_1^x \frac{1}{6}(t+1) dt, & 1 < x < 3 \\ 1, & x \geq 3 \end{cases}$$

$$\left[\int_1^2 f(x) dx \right. \\ \left. \int_1^2 \frac{1}{6}(x+1) dx \right. \\ \left. \text{or} \right. \\ \left. F(2) - F(1) \right] =$$

5) If $F(x)$ is the cumulative distribution of X random variable; find the probabilities;

$$F(x) = \begin{cases} 0, & x < 0 \\ x/4, & 0 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$



i) $P(X \leq -1) = F(-1) = 0$

ii) $P(X \leq 1) = F(1) = 1/4$

iii) $P(1 < X \leq 3) = F(3) - F(1) = \frac{3}{4} - \frac{1}{4}$

iv) $P(X > 0.75)$

$$= 1 - P(X \leq 0.75) = 1 - F(0.75)$$

$$= 1 - \frac{0.75}{4}$$

6) A loaded coin, the probability of showing tail is $\frac{2}{3}$ and showing head is $\frac{1}{3}$, is tossed three times. If random variable of X is the number of tails, what is the expected value of X ? $2^3 = 8$

$$S = \{ TTT, TTH, THT, HTT, THH, HTH, HHT, HHH \}$$

$$P(TTT) = \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} = \frac{8}{27}$$

$$P(THH) = \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{2}{27}$$

$$P(TTH) = \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{4}{27}$$

$$P(HTH) = \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{2}{27}$$

$$P(THT) = \frac{4}{27}$$

$$P(HHT) = \frac{2}{27}$$

$$P(HTT) = \frac{4}{27}$$

$$P(HHH) = \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{27}$$

$$R_x = \{ 0, 1, 2, 3 \}$$

$$E(x) = 0 \cdot \frac{1}{27} + 1 \cdot \frac{6}{27} + 2 \cdot \frac{12}{27} + 3 \cdot \frac{8}{27} = 2$$

7) Given the probability density function

$$f(X) = \begin{cases} \frac{4}{\pi(1+x^2)}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

of the random variable X , find the expected value and the standard deviation of X .

$$\begin{aligned} E(\underline{\hat{x}}) &= \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_0^1 x \cdot \frac{4}{\pi(1+x^2)} dx = \frac{4}{\pi} \int_0^1 \frac{x}{1+x^2} dx = \frac{4}{\pi} \int_1^2 \frac{1}{u} \frac{du}{2} \\ &= \frac{2}{\pi} \ln(1+x^2) \Big|_0^1 = \frac{2}{\pi} [\ln 2 - \ln 1] = \frac{\ln 4}{\pi} \end{aligned}$$

$\begin{matrix} 1+x^2=u \\ 2x dx = du \end{matrix}$

$$E(x^2) = \int_0^1 x^2 \cdot \frac{4}{\pi(1+x^2)} dx = \frac{4}{\pi} \int_0^1 \frac{x^2}{1+x^2} dx = \frac{4}{\pi} \int_0^1 \left(1 - \frac{1}{1+x^2}\right) dx$$

$$\frac{4}{\pi} \left(x - \arctan x \right) \Big|_0^1 = \frac{4}{\pi} \left[1 - (\arctan 1 - \arctan 0) \right] = \frac{4}{\pi} - 1$$

$\swarrow \pi/4$

$$\sigma^2 = E(x^2) - (E(x))^2 = \frac{4}{\pi} - 1 - \left(\frac{\ln 4}{\pi}\right)^2$$

8) A ball is randomly selected from a bag of N balls which are numerated from 1 to N . If X represents the numbers on the balls, find $E(X)$, $E(X^2)$ and σ_X^2 .

$X = x$	$f(x)$	$x \cdot f(x)$	$x^2 f(x)$
1	$1/N$	$1/N$	$1^2/N$
2	$1/N$	$2/N$	$2^2/N$
3	$1/N$	$3/N$	$3^2/N$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
N	$1/N$	N/N	N^2/N

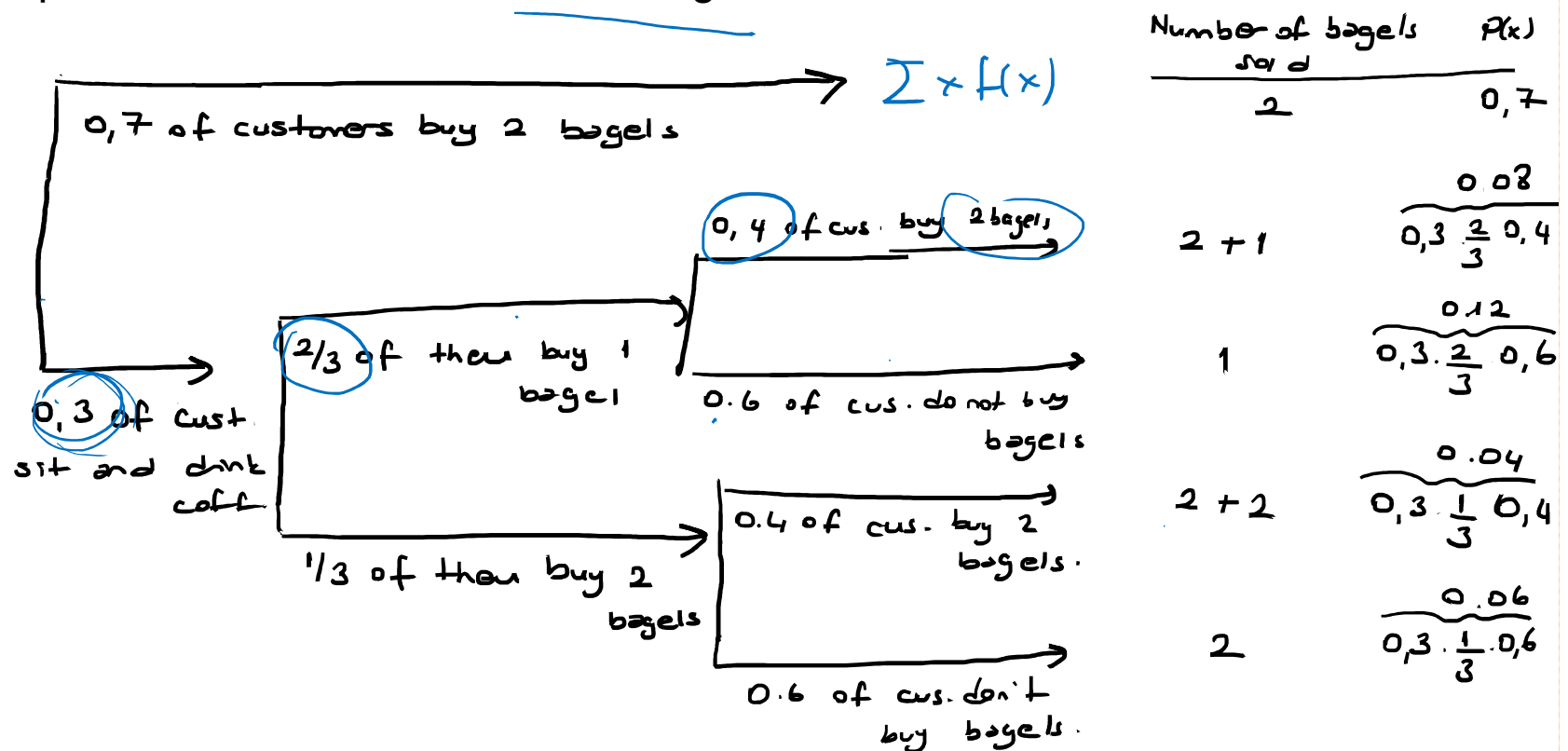
$$E(x) = \sum_{x=1}^N x f(x) = \frac{1}{N} (1 + 2 + 3 + \dots + N) = \frac{N+1}{2}$$

$\downarrow$
 $\frac{N(N+1)}{2}$

$$E(x^2) = \sum_{x=1}^N x^2 f(x) = \frac{1}{N} (1^2 + 2^2 + \dots + N^2) = \frac{(N+1)(2N+1)}{6}$$

$\downarrow$
 $\frac{N(N+1)(2N+1)}{6}$

9) 70% of the customers who come to a cake shop buy 2 bagels and leave the cake shop while 30% of them sit and drink coffee. $\frac{2}{3}$ of those who sit and have coffee take 1 bagel and have breakfast as well as $\frac{1}{3}$ of them take 2 bagels and have breakfast. 40% of the seats who have bought 1 bagel and 40% of the seats who have bought 2 bagel buy also 2 bagels when they leave. (60% of the seats do not buy bagels when they leave). Find the expected value of the number of bagels sold for each customer.



$$E(X) = 2 \times (0,7) + 3 \cdot (0,08) + 1 \cdot (0,12) + 4 \cdot (0,04) + 2 \cdot (0,06)$$

$$= 2,04$$

10) It is assumed that the probability to live 10 more years is 0.25 for Ahmet and is 0.40 for his wife.

- a) What is the probability that both of them live together for 10 more years?
- b) What is the probability that Ahmet or his wife live for 10 more years?
- c) What is the probability that none of them live for 10 more years?
- d) What is the probability that Ahmet not live for 10 more years but his wife live for 10 more years?

A : Ahmet lives 10 more years $P(A) = 0.25$

B : His wife " " $P(B) = 0.40$

$$a) P(A \cap B) = P(A)P(B) = (0.25)(0.40) = 0.1$$

$$b) P(A \cup B) = P(A) + P(B) - P(A \cap B) = (0.25) + (0.40) - (0.1)$$

$$c) P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B}) = (0.75)(0.60) = 0.45$$

$$d) P(\bar{A} \cap B) = P(\bar{A}) \cdot P(B) = (0.75)(0.40) = 0.30$$

11)

$$M_X(t) = e^{3t + 8t^2} \quad \checkmark$$

$$Z = \frac{1}{4}(X-3) \quad , \quad \underline{E(Z)} = ? \quad , \quad \underline{\sigma_Z^2} = ?$$

$$\underline{M_Z(t)} = M_{\frac{X-3}{4}}(t) = E\left[e^{(\frac{X-3}{4})t}\right] = e^{-\frac{3}{4}t} E\left[e^{\frac{X}{4}t}\right] = e^{-\frac{3}{4}t} \underline{M_X\left(\frac{t}{4}\right)}$$

$$M_Z(t) = e^{-\frac{3}{4}t} \cdot e^{3 \cdot \frac{t}{4} + 8 \cdot \left(\frac{t}{4}\right)^2} = e^{t^2/2} \quad \checkmark$$

$$E(Z) = \left. \frac{dM_Z(t)}{dt} \right|_{t=0} = \left. t \cdot e^{t^2/2} \right|_{t=0} = 0$$

$$E(Z^2) = \left. \frac{d^2 M_Z(t)}{dt^2} \right|_{t=0} = \left. e^{t^2/2} + t \cdot t \cdot e^{t^2/2} \right|_{t=0} = 1$$

$$\sigma_Z^2 = E(Z^2) - (E(Z))^2 = 1 - 0^2 = 1 \quad \checkmark$$

SOME IMPORTANT INEQUALITIES

1) Markov Inequality

Theorem: If X is a random variable that takes non-negative values and c is a constant number; then

$$P(X \geq c) \leq \frac{E(X)}{c}$$

Handwritten notes: A blue box surrounds the inequality. A blue arrow points from the text "This is Markov Inequality." to the box. To the right, a blue circle contains $f(x)$ with an equals sign below it, and an arrow points down to $E(x)$. Another blue arrow points from $E(x)$ to the $E(X)$ term in the denominator of the inequality.

This is Markov Inequality.

Ex: X is the random variable that shows the age of children selected randomly in a children's theater. Since the mean of this random variable is $E(X) = 5.5$, calculate the probability that a randomly selected child will be at least 17 by using the Markov inequality.

Solution:

$$P(X \geq \underline{17}) \leq \frac{5,5}{17} = \underline{0,32}$$

Handwritten notes: A blue box surrounds the equation. A blue checkmark is to the right of the equation. The number 17 in the denominator is underlined, and the result 0,32 is underlined with a wavy line.

2) Chebyshev Inequality

THEOREM 8. (*Chebyshev's Theorem*) If μ and σ are the mean and the standard deviation of a random variable X , then for any positive constant k the probability is *at least* $1 - \frac{1}{k^2}$ that X will take on a value within k standard deviations of the mean; symbolically,

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}, \quad \sigma \neq 0$$

$\Downarrow$

$$P(|x - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

$$P(|x - \mu| \geq \underbrace{k}_{\substack{\uparrow \\ \text{сб}}}) \leq \frac{\sigma^2}{k^2} \quad \checkmark$$

$\text{Var}(x) = \overline{(x - \mu)^2}$
 $E(x) = \mu$

Ex: Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50.

- a) What can be said about the probability that this week's production will be at least 75?
- b) If the variance of a week's production is known to equal 25, then what can be said about the probability that this week's production will be between 40 and 60?

Solution:

a) By Markov's inequality

$$P(X \geq c) \leq \frac{E(X)}{c}$$

$$P(X \geq 75) \leq \frac{E(X)}{75} = \frac{50}{75} = \frac{2}{3}$$

b) By Cheybshev's inequality

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

$$P(40 < X < 60) = P(40 - 50 < X - 50 < 60 - 50) = P(|X - 50| < 10)$$

$$\sigma^2 = 25 \Rightarrow \sigma = 5, \quad k\sigma = 10 \Rightarrow k \cdot 5 = 10 \Rightarrow k = 2$$

$$P(|X - 50| < 10) \geq 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} = 0,75$$

Example: In a class, students have IQ scores with the mean 70 and standard deviation 4; At least, what percentage of the students have IQ scores between 58 and 82?

Solution: By using Chebyshev's theorem

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

$$P(58 < x < 82) = P(58 - 70 < x - 70 < 82 - 70)$$

$$= P(|x - 70| < 12)$$

$$k\sigma = 12 \Rightarrow k.(4) = 12$$

$$\Rightarrow k = 3$$

$$k = 3$$



$$1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} = 0,89 = \%89$$

Example: TV tube production is carried out in a factory. Assuming mean of the weekly production is 1500

- a) what is the probability that this week's television tube production is at least 2000?
- b) If $\sigma^2=1$, then what is the probability that the production will be between 1400 and 1600 in the same week?

Solution: a) By using Markov inequality;

$$P(X \geq c) \leq \frac{E(X)}{c}$$

$$P(X \geq 2000) \leq \frac{E(X)}{2000} = \frac{1500}{2000} = 0,75$$

b) By using Chebyshev's theorem;

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

$$\begin{aligned} P(1400 < x < 1600) &= P(1400 - 1500 < x - 1500 < 1600 - 1500) \\ &= P(|x - 1500| < 100) \end{aligned}$$

$$k\sigma = 100 \Rightarrow k \cdot (1) = 100 \quad \Rightarrow \quad k = 100$$

$$P\{|X - 1500| < 100\} \geq 1 - \frac{1}{100^2} = 0,99$$

Example: The mean and variance of the weekly cost of a small shop is 1000 TL and 400 TL, respectively. Determine an interval for the weekly cost whose occurring probability is at least 75%.

$$\mu = 1000, \sigma^2 = 400 \Rightarrow \sigma = 20$$

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \quad 1 - \frac{1}{k^2} = 0.75 \Rightarrow k = 2$$

$$P(|x - 1000| < 2 \times 20) \geq 0.75$$

$$P(|x - 1000| < 40) \geq 0.75 \Rightarrow P(-40 < x - 1000 < 40) \geq 0.75$$

$$P(\underline{960} < x < \underline{1040}) \geq 0.75$$

Ex: X is the random variable that shows the age of children selected randomly in a children's theater. Since the average of this random variable is 5.5 and standard deviation is 1, calculate the probability that a randomly selected child will be at least 17.

$$\mu = E(x) = 5.5, \sigma = 1 \quad P(x \geq 17) = ?$$

$$P(x - \mu \geq \underset{\substack{\downarrow \\ 5.5}}{17 - \mu}) = P(|x - \mu| \geq 11.5) \leq \frac{\sigma^2}{(11.5)^2} = \frac{1}{(11.5)^2} = 0.0076$$